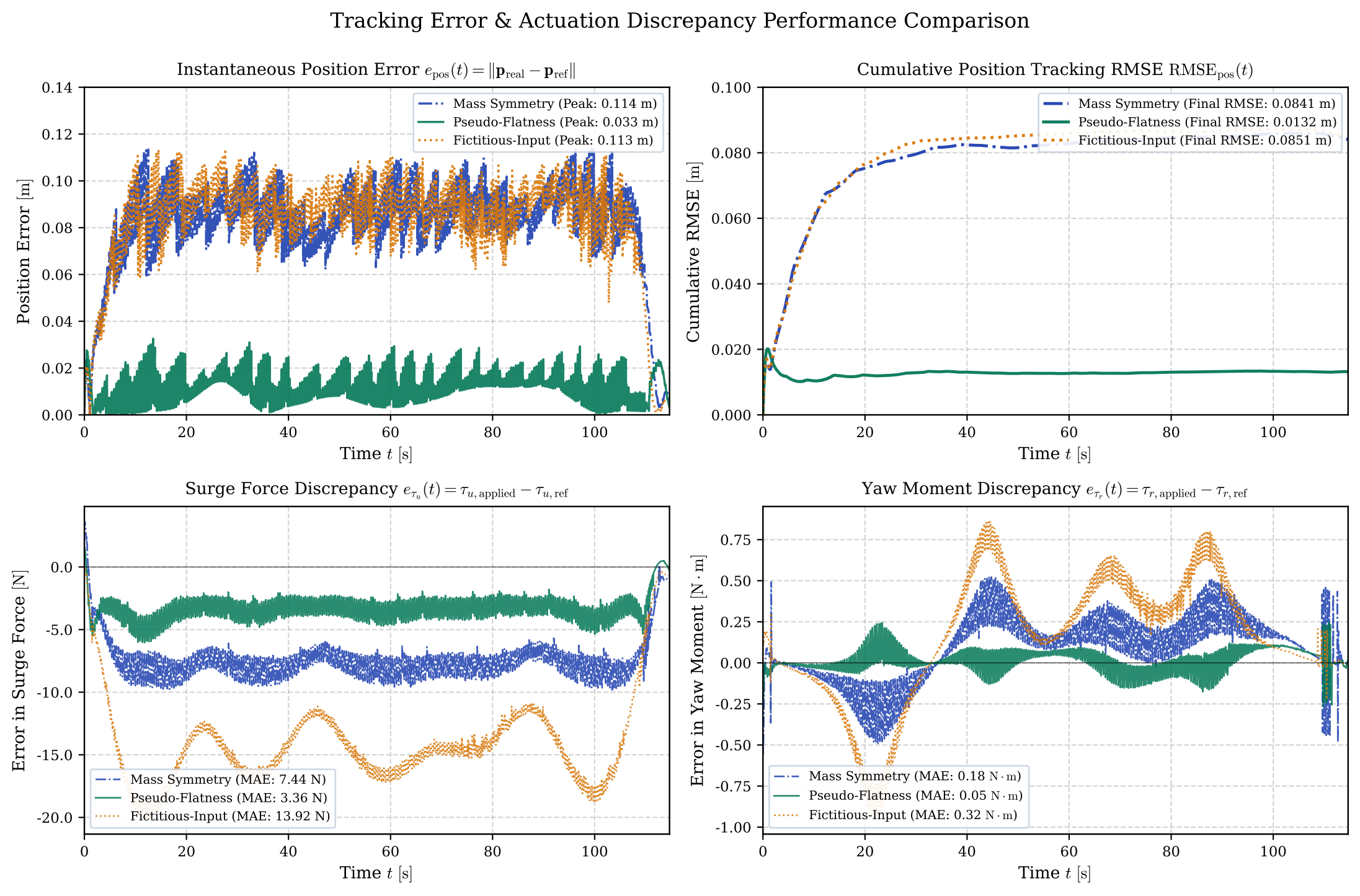

Tracking Error & Actuation Discrepancy Performance Comparison
Instantaneous Position Error $e_{\text{pos}}(t) = \|\mathbf{p}_{\text{real}} - \mathbf{p}_{\text{ref}}\|$
Position Error [m]
Time t [s]
Mass Symmetry (Peak: 0.114 m)
Pseudo-Flatness (Peak: 0.033 m)
Fictitious-Input (Peak: 0.113 m)
Cumulative Position Tracking RMSE $\text{RMSE}_{\text{pos}}(t)$
Cumulative RMSE [m]
Time t [s]
Mass Symmetry (Final RMSE: 0.0841 m)
Pseudo-Flatness (Final RMSE: 0.0132 m)
Fictitious-Input (Final RMSE: 0.0851 m)
Surge Force Discrepancy $e_{\tau_u}(t) = \tau_{u,\text{applied}} - \tau_{u,\text{ref}}$
Error in Surge Force [N]
Time t [s]
Mass Symmetry (MAE: 7.44 N)
Pseudo-Flatness (MAE: 3.36 N)
Fictitious-Input (MAE: 13.92 N)
Yaw Moment Discrepancy $e_{\tau_r}(t) = \tau_{r,\text{applied}} - \tau_{r,\text{ref}}$
Error in Yaw Moment [N · m]
Time t [s]
Mass Symmetry (MAE: 0.18 N · m)
Pseudo-Flatness (MAE: 0.05 N · m)
Fictitious-Input (MAE: 0.32 N · m)